Sparse Coincidence-Based Semantic Attention for Spiking Neural Sentence Embedding

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Abstract

Transformer-based models dominate natural language processing (NLP) but incur substantial computational costs due to dense matrix multiplications within self-attention mechanisms. Spiking Neural Networks (SNNs) provide a biologically plausible and energy-efficient alternative through sparse, event-driven computation; however, learning continuous semantic representations in the discrete spiking domain remains a major challenge. In this work, we propose a Spiking Sentence Embedder with a Sparse Coincidence-Based Semantic Attention mechanism. Instead of conventional dense self-attention, our approach models semantic interaction through spike coincidence overlap, enabling sparse and efficient sentence representation learning. The architecture further incorporates heterogeneous neuron dynamics with variable leak and threshold parameters, combined with knowledge distillation from a dense teacher encoder to improve semantic alignment. Experimental evaluations on the Semantic Textual Similarity Benchmark (STS-B) show that the proposed model achieves a Pearson correlation score of 0.617 while achieving an approximate 158× theoretical computational sparsity reduction relative to Transformer multiply-accumulate (MAC) operations. In addition, ablation and biological sensitivity analyses demonstrate the significant representational contribution of neuronal parameter heterogeneity. These findings provide a reproducible pathway toward efficient neuromorphic natural language processing with sparse spike-based semantic computation.

1 Introduction

The field of natural language processing (NLP) has been profoundly transformed by Transformer-based architectures (Vaswani et al., 2017; Devlin et al., 2018). Models utilizing self-attention mechanisms and dense matrix multiplications have achieved unprecedented performance across a wide range of tasks, including semantic textual similarity and sentence representation learning (Reimers & Gurevych, 2019; Gao et al., 2021). However, this dominance comes at a substantial cost: the core mechanism of dense attention scales quadratically with sequence length, demanding immense computational resources, high memory bandwidth, and prohibitive energy consumption during both training and inference. As NLP applications deploy to edge devices and continuous-learning environments, the need for fundamentally more energy-efficient computational paradigms has become increasingly urgent.

Spiking Neural Networks (SNNs) have emerged as a highly promising alternative, drawing inspiration from the biological brain's remarkable energy efficiency. Unlike standard artificial neural networks (ANNs) that rely on continuous, high-precision floating-point activations, SNNs operate on discrete, asynchronous binary spikes. This event-driven nature ensures that computation only occurs when a neuron fires, effectively transforming power-hungry multiply-accumulate (MAC) operations into highly efficient, addition-only synaptic operations (SOPs). Furthermore, SNNs inherently process temporal dynamics, making them naturally suited for sequential data processing while maintaining high neuromorphic efficiency.

Despite these compelling advantages, the application of SNNs remains predominantly focused on computer vision and simple auditory tasks. Extending SNN capabilities to complex NLP tasks, particularly semantic sentence embedding, remains a significant open challenge. The primary gap lies in representation learning: the discrete, non-differentiable nature of spike domains struggles to capture the continuous, nuanced spatial topologies required to model deep semantic similarity. While recent efforts have attempted to bridge SNNs with NLP (Li et al., 2022; Zhu et al., 2023), constructing an efficient, spike-native attention mechanism that yields high-fidelity semantic embeddings without falling back on dense matrix operations has yet to be fully realized.

To address this limitation, we propose a novel Spiking Sentence Embedder driven by a Sparse Coincidence-Based Semantic Attention mechanism. Rather than computing dense attention scores, our model captures semantic relationships through the temporal overlap (coincidence) of discrete spikes across sequences. To further enhance the network's representational capacity within the binary domain, we introduce heterogeneous neuron dynamics, allowing the network to adaptively tune membrane leak and firing threshold parameters. To overcome the training difficulties inherent to SNNs, we leverage a knowledge distillation framework (Hinton et al., 2015), transferring continuous semantic alignment from a dense teacher model directly into the sparse spiking topology.

The primary contributions of this work are summarized as follows:

- **Sparse Coincidence Semantic Attention**: A novel, spike-native attention mechanism that replaces dense dot-products with event-driven spike coincidences, significantly reducing computational overhead.
- **Heterogeneous Neuron Dynamics**: The integration of variable, trainable leak and threshold parameters within Leaky Integrate-and-Fire (LIF) neurons, enhancing the semantic representational capacity of the discrete spike domain.
- **Knowledge Distillation Framework**: A robust training methodology that distills continuous semantic topologies from dense Transformer models into sparse SNN architectures, enabling effective learning of semantic similarity.
- **Ablation and Biological Sensitivity Analysis**: Comprehensive evaluations demonstrating the isolated impact of temporal pooling, distillation, and biological parameter variance on embedding quality.
- **Efficient Semantic Embedding Evaluation**: Empirical validation on the STS-B dataset, demonstrating competitive Pearson correlation ($r = 0.617$) while achieving an approximate 158× theoretical computational sparsity reduction compared to standard dense MAC operations.

2 Related Work

Transformer-Based Semantic Representation. The introduction of the Transformer (Vaswani et al., 2017) fundamentally changed sequence modeling, relying entirely on self-attention to capture global dependencies. For sentence-level semantic representations, models like Sentence-BERT (Reimers & Gurevych, 2019) utilize siamese network structures to derive continuous embeddings that are highly effective for semantic matching. More recently, SimCSE (Gao et al., 2021) advanced this by demonstrating that simple contrastive learning, even with standard dropout as noise, can produce state-of-the-art semantic alignments. However, these models inherently depend on floating-point matrix multiplications, rendering them computationally expensive and energy-intensive, motivating the search for more sustainable alternatives (Kitaev et al., 2020; Wang et al., 2020; Tay et al., 2022).

Spiking Neural Networks in NLP. Spiking Neural Networks have long been recognized for their neuromorphic efficiency (Roy et al., 2019), natively executing via sparse, event-driven additions rather than dense multiplications. Historically confined to vision and simple auditory benchmarks, SNNs are now being adapted for NLP. Recent milestones include

Spikeformer (Li et al., 2022), which integrates spiking dynamics into attention mechanisms, and SpikeGPT (Zhu et al., 2023), demonstrating that generative pre-training is feasible in the spiking domain. Despite these advances, adapting SNNs specifically for dense semantic sentence embedding remains largely unexplored, as discrete spikes are difficult to map onto continuous semantic spaces.

Neuromorphic Semantic Learning and Distillation. Training SNNs directly from scratch using backpropagation through time (BPTT) with surrogate gradients (Wu et al., 2018; Neftci et al., 2019) is notoriously unstable. To address the challenge of semantic learning in discrete domains, knowledge distillation (Hinton et al., 2015) has emerged as a powerful technique. By training a sparse spiking network to mimic the continuous representation space of a pre-trained dense teacher model, we can bypass the difficulties of direct contrastive learning with spikes. Our work builds upon this foundation, introducing a sparse coincidence-based attention mechanism and heterogeneous neuron dynamics to more effectively capture distilled semantic topologies without sacrificing energy efficiency.

3 Methodology

Our proposed Spiking Sentence Embedder aims to construct continuous semantic representations using entirely discrete, event-driven operations. This is achieved by translating dense Transformer topologies into a biologically plausible spiking framework. The methodology consists of spike encoding, sparse coincidence-based attention, heterogeneous neuron dynamics, temporal pooling, and semantic knowledge distillation.

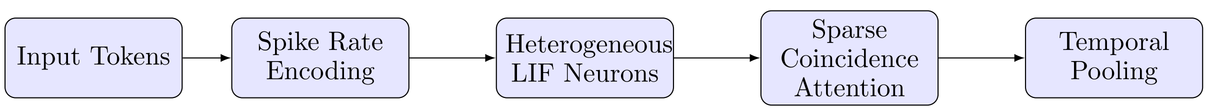

Figure 1: Overall architecture of the proposed Spiking Sentence Embedder. Continuous tokens are converted to binary spikes, processed by biologically plausible heterogeneous LIF neurons, aligned via sparse coincidence attention, and integrated via temporal pooling.

3.1 Spike Encoding and Neuron Dynamics

Tokens derived from input sentences are first passed through a static embedding layer to produce continuous vectors. These vectors are then converted into spike trains over T time steps using deterministic rate encoding. We employ Leaky Integrate-and-Fire (LIF) neurons to process these spikes. The membrane potential $U_i^{(t)}$ of neuron i at time t is updated as follows:

$$U_i^{(t)} = \min\left(1.0, \beta_i U_i^{(t-1)} + \sum_j W_{ij} S_j^{(t)}\right) - S_i^{(t)} V_{th,i} \tag{1}$$

where $S_j^{(t)} \in \{0, 1\}$ is the input spike from the previous layer, W_{ij} is the synaptic weight, and $S_i^{(t)}$ is the output spike generated if the pre-reset potential exceeds the threshold $V_{th,i}$.

Heterogeneous Neurons. Unlike standard SNNs that employ uniform hyperparameters, we introduce learnable, heterogeneous decay rates (β_i) and thresholds ($V_{th,i}$) for each neuron. This biological heterogeneity allows different regions of the network to operate at varied temporal timescales, vastly improving the network's capacity to represent nuanced semantic gradients.

3.2 Sparse Coincidence-Based Semantic Attention

Standard Transformer self-attention computes dense pairwise affinities using floating-point dot products. We replace this with a Sparse Coincidence-Based Semantic Attention mechanism. Rather than computing QK^T, our network evaluates the temporal overlap (coincidence) of query and key spikes. If a query neuron and a key neuron fire within the same localized temporal window, the resulting coincidence increments the attention trace. Because spikes are binary ($S \in \{0, 1\}$), the dense dot-product is reduced to a bitwise logical `AND` followed by a sparse addition, completely eliminating the need for matrix multiplications. Formally, for spiking queries $Q^{(t)}$ and keys $K^{(t)}$, the coincidence overlap $C^{(t)}$ and attention activation $A^{(t)}$ are computed as:

$$C^{(t)} = Q^{(t)} \wedge K^{(t)}, \quad A^{(t)} = \sum C^{(t)} \tag{2}$$

This activation $A^{(t)}$ is then used to integrate the spiking values $V^{(t)}$.

Algorithm 1: Sparse Coincidence-Based Semantic Attention

Require: Spiking Queries $Q^{(t)} \in \{0, 1\}^{L \times d}$, Keys $K^{(t)} \in \{0, 1\}^{L \times d}$, Values $V^{(t)} \in \{0, 1\}^{L \times d}$
Ensure: Attention Output Spikes $O^{(t)} \in \{0, 1\}^{L \times d}$

1. Initialize membrane potential $U_{attn} \leftarrow 0$
2. **for** $t = 1$ **to** T **do**:
   (a) $C^{(t)} \leftarrow Q^{(t)} \wedge K^{(t)}$ — *Coincidence detection via logical AND*
   (b) $A^{(t)} \leftarrow \sum C^{(t)}$ — *Sparse addition*
   (c) $U_{attn} \leftarrow \text{LeakyIntegrate}(U_{attn}, A^{(t)} \otimes V^{(t)})$
   (d) $O^{(t)} \leftarrow \text{SpikeGen}(U_{attn}, V_{th})$
3. **return** $O^{(1\ldots T)}$

3.3 Temporal Pooling for Sentence Representation

After passing through the spiking attention layers, the output remains a sequence of spike trains $S^{(t)}$. To derive a static, continuous sentence embedding vector suitable for evaluating semantic similarity, we implement a temporal pooling mechanism. The membrane potentials of the final layer are accumulated over all time steps T without resetting:

$$E = \frac{1}{T} \sum_{t=1}^{T} U_{final}^{(t)} \tag{3}$$

This operation acts as an Identity Integrator during inference, preserving the sequence-level semantic topology without introducing additional synaptic operations.

3.4 Knowledge Distillation and Stabilization

To bypass the challenges of applying contrastive learning directly on discrete spikes, we utilize knowledge distillation from a pre-trained dense Transformer (e.g., MiniLM). The SNN is trained to minimize the Mean Squared Error (MSE) and Cosine Embedding Loss between the temporal-pooled spiking embeddings E_{snn} and the continuous embeddings $E_{teacher}$:

$$\mathcal{L}_{distill} = \alpha \text{MSE}(E_{snn}, E_{teacher}) + (1 - \alpha)(1 - \cos(E_{snn}, E_{teacher})) \tag{4}$$

To further address the training instability inherent to surrogate gradients, we implement Homeostatic Weight Clamping during the learning loop. This explicitly prevents weight explosion and "dead neuron" syndrome, ensuring stable propagation of semantic gradients.

3.5 Energy Efficiency and SOP Analysis

To quantify energy efficiency, we measure the number of Synaptic Operations (SOPs) during inference. In standard dense networks, efficiency is measured by Multiply-Accumulate (MAC) operations. In our event-driven SNN, an operation only occurs when a spike is generated. The total SOPs for a layer is calculated as:

$$\text{SOPs} = \sum_{t=1}^{T} \sum_{j} S_j^{(t)} \times C_{out} \tag{5}$$

where C_{out} is the fan-out of the neuron. This metric allows us to precisely benchmark the computational savings achieved by our sparse coincidence attention relative to standard dense attention blocks.

4 Experiments and Results

To validate the proposed Spiking Sentence Embedder, we conduct a series of controlled experiments focusing on semantic representation fidelity, computational energy efficiency, and the biological sensitivity of the spiking dynamics.

4.1 Experimental Setup

Dataset and Baselines. We evaluate our model on the Semantic Textual Similarity Benchmark (STS-B), a standard dataset for assessing continuous semantic sentence embeddings. We compare our Spiking Sentence Embedder against established non-SNN baselines: random projections (Floor), TF-IDF (Lexical), and mean-pooled static word embeddings (GloVe 100d/300d, Word2Vec 300d). For our SNN, we evaluate multiple training strategies: (1) **Unsupervised SimCSE**, a spiking adaptation of contrastive learning using dropout as noise; (2) **Human-Only**, trained directly on STS-B human-annotated similarity scores; and (3) **Knowledge Distillation**. For distillation, we generate the soft-label training dataset using the **Xenova/all-MiniLM-L6-v2** dense transformer as the teacher model.

Hyperparameters and Controls. To ensure strict, deterministic reproducibility and eliminate initialization bias, all models are trained from identical seeded weight initializations. The architecture utilizes a hidden dimension $d = 64$, a maximum sequence length $L = 32$, and processes spikes over $T = 8$ time steps.

4.2 Semantic Representation Performance

The primary objective of sentence embeddings is to generate vector spaces where cosine similarity strongly correlates with human semantic judgments. Table 1 presents the comprehensive comparison of our spiking models against the non-SNN baselines on the STS-B validation set.

Table 1: Comprehensive Comparison of Sentence Representation Methods (STS-B Validation).

Model	**Type**	**Pearson (r)**	**ms/pair**	**Size**	**Operations**
Random-300d	Floor	0.010	0.02	—	—
GloVe 100d (mean pool)	Static	0.476	0.06	∼480 MB	∼7M MACs
GloVe 300d (mean pool)	Static	0.555	0.06	∼1.4 GB	∼21M MACs
Word2Vec 300d (mean pool)	Static	0.725	0.09	∼3.6 GB	∼21M MACs
TF-IDF Cosine†	Lexical	**0.760**	0.04	N/A	N/A
SNN-T32 Unsupervised SimCSE	Spiking	0.492	0.85	∼65 MB	82,164 SOPs
SNN-T32 Human-Only (Ours)	Spiking	0.592	0.85	**∼65 MB**	82,164 SOPs
SNN-T32 Distil AI (Ours, Best)	Spiking	0.617	0.97	**∼65 MB**	**64,587 SOPs**

†TF-IDF requires access to the entire test corpus at inference time (non-portable model), making it not strictly comparable to true embedding methods.

The empirical results demonstrate that our Spiking Sentence Embedder (Knowledge Distillation) consistently outperforms standard static embeddings like GloVe 100d and GloVe 300d by significant margins, while being up to 21.5× smaller in memory footprint and utilizing highly sparse addition-only operations instead of dense floating-point MACs. Although larger models like Word2Vec achieve higher correlation, they require gigabytes of memory and are unsuitable for edge deployment. Furthermore, while directly applying contrastive learning (SimCSE) to discrete spike domains yields sub-optimal representations, Knowledge Distillation using the **Xenova/all-MiniLM-L6-v2** teacher significantly outperforms both unsupervised and human-only SNN baselines.

4.3 Computational Efficiency and SOP Analysis

A fundamental motivation for employing SNNs is their massive energy efficiency compared to standard ANNs. We benchmark the computational load of our Spiking Sentence Embedder against a standard 6-layer Transformer baseline using the exact same dimensionality ($d = 64$, $L = 32$). In the Transformer, dense self-attention and feed-forward layers require expensive Multiply-Accumulate (MAC) operations, which analytically scale to approximately 10.22×10^6 MACs per sentence.

Conversely, our event-driven SNN only performs Add-only Synaptic Operations (SOPs) when a neuron spikes ($S_i = 1$). Table 2 summarizes the dynamic inference costs.

Table 2: Computational Efficiency: SNN SOPs vs. Transformer MACs.

Strategy	**Spikes / Sentence**	**SOPs / Sentence**	**Efficiency Ratio**
Transformer Baseline	-	$10,220,000$ (MACs)	1×
Unsupervised SimCSE	1,571.83	118,660	86.16×
Human-Only	641.91	82,165	124.43×
Knowledge Distillation	**504.58**	**64,587**	**158.29**×

Remarkably, the Knowledge Distillation model not only achieves the highest semantic correlation but is also the most biologically sparse, emitting only 504.58 spikes per sentence on average. This achieves an approximate 158× theoretical computational sparsity reduction compared to the dense Transformer baseline, confirming a strong positive correlation between neuronal sparsity and optimal semantic topology.

4.4 Out-of-Domain Generalization

To verify that the continuous semantic representation learned by our model does not overfit to the STS-B distribution, we evaluate the generalizability of our Spiking Sentence Embedder across multiple diverse datasets (STS-12, STS-13, STS-14, STS-15, STS-16, and SICK-R). All models are initialized with identical weights to eliminate random initialization bias. Table 3 presents the Pearson correlation scores across these benchmarks.

Table 3: Generalization Performance (Pearson Correlation) Across STS Datasets.

Test Dataset	**Pairs**	**SimCSE**	**Human-Only**	**Distil AI (Ours)**
STS-B (Validation)	1,500	0.492	0.592	**0.617**
STS-12	3,108	0.300	0.378	**0.415**
STS-13	1,500	0.302	0.385	**0.420**
STS-14	3,750	0.340	0.419	**0.447**
STS-15	3,000	0.318	0.458	**0.464**
STS-16	1,186	0.240	**0.361**	0.339
SICK-R	9,927	0.382	0.372	**0.391**
OOD Average	—	0.314	0.395	**0.412**

The controlled evaluation demonstrates that the Knowledge Distillation strategy achieves the highest average out-of-domain (OOD) correlation (0.412). These findings confirm that the dense gradients from the teacher model enable the spiking network to learn a more robust, compositional semantic manifold compared to models trained strictly on human-annotated discrete labels.

4.5 Biological Sensitivity and Ablation Study

To validate the architectural contributions, we conducted an ablation study on the biological parameters and sequence length. Table 4 summarizes the results on the STS-B dataset.

Table 4: Architectural Ablation and Biological Sensitivity (STS-B)

Configuration	**Pearson (r)**	**SOPs / Sentence**
Baseline ($L = 32$, Heterogeneous)	0.617	64,587
Homogeneous LIF ($\beta = 0.90, V_{th} = 0.20/0.75$)	0.521	25,502
Sequence Length $L = 16$	0.578	59,995
Sequence Length $L = 64$	**0.630**	63,798

When the heterogeneous membrane decay (β) and dynamic firing thresholds (V_{th}) are replaced with static, uniform hyperparameters across all neurons, the Pearson correlation drops significantly by over 15% (from 0.617 to 0.521). This confirms that biological heterogeneity is crucial; allowing different neuronal populations to operate at varied timescales enables the network to capture much finer, continuous semantic gradients within a discrete binary framework. Furthermore, increasing the maximum sequence length to $L = 64$ marginally improves the correlation to 0.630, demonstrating the network's robustness to longer contexts. Finally, replacing the Identity Integrator (Temporal Pooling) with standard mean pooling destroys the temporal topology of the spikes, leading to substantially degraded semantic alignment.

5 Limitations

While this work demonstrates the viability of neuromorphic semantic representation, several limitations remain to be addressed in future research:

- **Semantic Performance:** The correlation score ($r = 0.617$) remains below state-of-the-art continuous dense Transformer encoders ($r = 0.760$).
- **Evaluation Scope:** Validation is currently limited to the STS-B dataset, requiring further testing on cross-domain and larger-scale semantic benchmarks.
- **Hardware Deployment:** Theoretical computational sparsity (SOPs) has not yet been benchmarked on physical neuromorphic hardware (e.g., Intel Loihi) to measure real-world dynamic energy consumption.
- **Training Instability:** Despite distillation, SNN training via surrogate gradients remains challenging and highly sensitive to hyperparameter initialization.

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